STANDARD

RENAULT

01 - 33 - 200 / - - D

DESIGN AND PROCESS FMEA

(FAILURE MODE, EFFECTS

AND CRITICALITY ANALYSIS)

Renault Automobiles Standardisation
DQSC-D / Department 00624

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FIRST ISSUE

December 1986	- - -

REVISIONS

November 2000	- - A	Complete revision. This issue originates from draft NC 1999 0171 / - - B
December 2009	- - B	Adjusting the standard so that it meets market requirements and Company requirements. • Contribution of information to definitions and method in which risks are analysed and criticality is evaluated. • Update of evaluation criteria related to Severity, Occurrence and Design Control. • Update of RPN thresholds. • Addition of diagnosis grid to Appendices. This issue originates from draft NC 2007 0704 / - - -
January 2011	- - C	Conformation with the new V3P milestones This issue originates from draft NC 2011 0084 / - - -
December 2019	- - D	Taking into account the release of new FMEA International reference applied to the automotive industry "AIAG / VDA FMEA HandBook" -Update severity grid: Small non-disabling injuries in Severity 6 -Update initialization sheet -Creation of FMEA Product & Process Template (appendix 3) -Creation of FMEA Suppliers Synthesis Template (appendix 7) -Adding of Safety and Regulation Special Characteristic concept

REFERENCED DOCUMENTS

CNOMO Standard	:	E41.50.530.N.
Renault Standard	:	Standard 00 - 10 - 040 : DEFINITION, IDENTIFICATION, OBLIGATIONS AND TRACKING OF PRODUCTS AND SAFETY OR REGULATORY CHARACTERISTICS
RPIFF (AER)	:	RPIFFPV3P20110137_ AWP-B-20099_: AMDE système (Analyse des Modes de Défaillances et de leurs Effets systèmes) RPIFFSGPR20150009-V4_AEK-D-00051_v2.0 : Liste EICps RN

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GLOSSARY

ABVC : Agreement to Built Vehicle Check (Vehicle Engineering ; V3P logic).

ABPT : Agreement to Built Plant Trial (Mechanical Engineering ; JPOM5 logic).

AMS-lot : S-Lot Assembly Approval (Vehicle Engineering ; E=MC² logic).

AMEI : Approval to Build Initial Samples (Mechanical Engineering ; JPOM4 logic).

AMIOD : Approval to Build Definitive Off-Tool Parts (Mechanical Engineering ; JPOM4 logic).

DLI : Industrial Logistics Department.

DPLI : Manufacturing and Industrial Logistics Department.

Durability: In the dependability area, durability is the capacity of equipment to perform a function until a limit state is reached (Standard NF X 60-500).

E ‰ : is the result indicator of the 3-month Estel survey that polls customer / users of the product.

FMEA : Failure Mode, Effects and Criticality Analysis (FMEA).

Functional Specifications : Functional Specifications.

IOD : Definitive Off-tool Part (Mechanical Engineering ; JPOM4 logic).

K‰ : Warranty cases per thousand.

NCR : Non-Conformity Rate.

RPN : Risk Priority Number.

PIMOL : Vehicle off-road breakdown.

PC : Pre-Contract (Mechanical Engineering ; JPOM5 logic).

PPM : Parts per Million (1 % = 10 K ‰ = 10 000 PPM).

RO : Tooling Go Ahead (Vehicle Engineering ; E=MC² logic).

SdF : Operating Dependability.

S1 ‰ : SOFRES per thousand after the customer has used the product for 1 year.

S3 ‰ : SOFRES per thousand after the customer has used the product for 3 years.

TGA : Tooling Go Ahead (V3P and JPOM5 logics).

VPC : Vehicle Pre-Contract (Vehicle Engineering ; V3P logic).

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0. LANGUAGE

The FMEAs that are compiled for the RENAULT Group must be in French (as a priority) or in English.

1. SCOPE AND FIELD OF APPLICATION

Failure Mode, Effects and Criticality Analysis (FMEA) is a method used to analyse the potential reliability of a product. This method contributes to the performance of the company.

FMEA **favours spending time and cost on prevention so that less will be spent during correction.**

The FMEA is used to identify the potential failures and related consequences that would affect the correct operation of the product, manufacturing process, production facilities or equipment being studied. It assess the risks linked to the occurrence of such failures so that corrective actions can be taken, if needed, at the design, implementation or operational (production, maintenance, monitoring) phase of the product, manufacturing process, facilities or equipment.

This standard deals with Design FMEA and Process FMEA.

The FMEA applies to the AUTOMOBILE PRODUCT at the different design stages of the Engineering Department. The scope of the study by system and/or element may cover the following:

- mechanical, hydraulic, electrical, mechatronics technology
- one unit, subset, component, characteristic of the product
- one main function and/or constraint with related value criteria

Note: The FMEA is not the ideal tool for evaluating software.

The FMEA applies to the industrialisation PROCESS of the product used by Production Engineering departments. The scope of the study by system and/or element may cover the following:

- production process, rework process or inspection process of a product,
- manufacturing operation and/or inter-operation,
- one or n manufactured characteristics, etc.

The FMEA also applies to:

- PRODUCTION FACILITIES during the phase in which the facilities are designed (see Machinery FMEA – CNOMO Standard E41.50.530.N).
- Physical and facilities WORKFLOWS during the phase in which operational modes are drawn up (see § 5. ADDITIONAL INFORMATION: FLOW FMEA in this standard).

In the context of projects, FMEA progress and the results obtained are decision-making elements that the Quality Department takes into consideration before issuing RECOMMENDATIONS and APPROVALS at Project milestones ("Vehicle" or "Mechanical Unit").

It is the responsibility of the Technical Departments and Suppliers to assume ownership of this standard.

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2. DEFINITIONS

FMEA

Failure Mode, Effects and criticality Analysis is a rigorous and preventive method that aims at listing and then evaluating the potential failures of a system and/or element. This method of analysis leads to the organisation of these failures into a hierarchy based on which the appropriate actions to be taken can be decided. A schedule that is used to follow up these corrective or monitoring actions is set up. Application of these measures is checked in the field.

The knowledge gained from every study is used as a diagnosis support and for future projects.

DESIGN FMEA

The scope of the analysis is the design and definition of the automobile product, which is examined in the various stages of its development before the Part Tooling Development (TGA) stage. This is undertaken in order to ensure that the technical solution decided upon meets specifications related to quantified requirements as well as customer expectations and industrial constraints.

Consequences of potential failures must be related to the customer / user of the product (driver, passenger, another road user, repair network, emergency organisations, recycler, etc.; it may also relate to regulations).

PROCESS FMEA

The scope of the analysis is designing the industrialisation plan (production process, assembly process, rework process or inspection process) of an automobile product, which is examined during the different planned operations of production: manufacturing, inspection, handling, etc. to ensure that as per the industrial process a product can be mass-produced in compliance with requirements specified in plans.

Improvement may concern either the product or the process, or both.

Consequences of potential failures must be related to the customer / user of the product.

This standard is also used to indicate the effect that a single type of product non-compliance may have on manufacturing/assembly, but without obligation (a Process FMEA, in which customer-side manufacturing/assembly effects are indicated, would not replace a facilities or flow FMEA).

PRODUCTION FACILITY FMEA

The scope of the analysis is the production tool, machine, robot, machining fixture, tooling, etc., which is examined during the design phase in order to ensure that during operation it will meet objectives related to product compliance as well as Reliability, Maintainability, Availability, Safety of people and equipment (FMDS), required in the specifications of the facility.

Consequences of potential failure are relative to the user of the facility: that is, the Manufacturer of the automobile product.

Note: The production facility FMEA is carried out in compliance with CNOMO standard E41.50.530.N. It has not been given in detail in this standard.

FLOW FMEA: see additional information in Appendix 5

The scope of the analysis is relative to the movement of parts of the product, its packaging and the flow of information while the facilities are in operation.

Consequences of potential failures are relative to the Manufacturer and/or Operator as well as the Industrial Logistics Department.

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System FMEA:

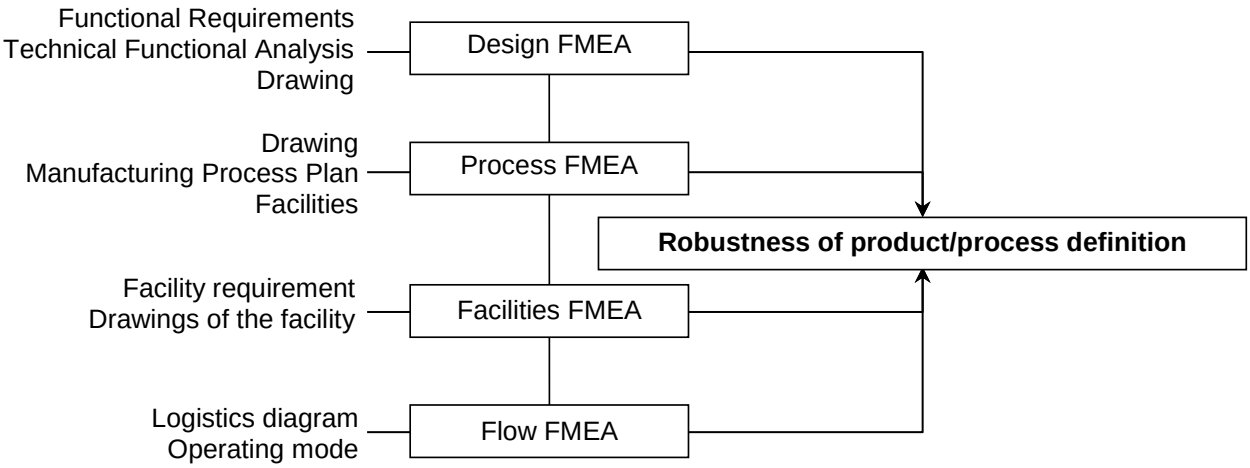
System architecture design. It mainly allows to validate the system architectures and to ensure that the user client cannot be affected by failures presenting in particular safety, regulatory and immobilizing breakdown risks.

Electronic FMEA:

Electronic Design FMEA of ECUs allows to understand the diagnosability. The Electronic FMEA is carried out in particular within the framework of the coverage of the metrics requested by ISO 26262 standard.

The FMEA is one of the Operating Reliability tools (see RPIFMETHQ20110002)

Implementing Product, Process, Facilities and Flow FMEA optimises the robustness of the product-process design on a priority basis and sets up, if needed, a monitoring plan that is adapted to risks.



3. BASIC PRINCIPLE

1) The FMEA is available to every one of us to construct quality and reliability measures that prevent failures, to meet customer / user requirements and to contribute to the performance of the industrial system.

In the context of **Customer – Supplier relations**, every person involved has a task to accomplish, either on an individual basis or collectively:

. To accomplish task "n", quantified input data from the Supplier who has completed upstream task "n-1" is necessary.

. While exiting this phase, we must provide results that are in compliance with the objective so that the person responsible for downstream task "n+1" can complete it.

The FMEA is a tool used to construct quality and reliability that contributes to the completion of task "n". **By implementing the FMEA, it is possible to validate the task or optimise it before sending it to "its direct Customer"** (figure 1).

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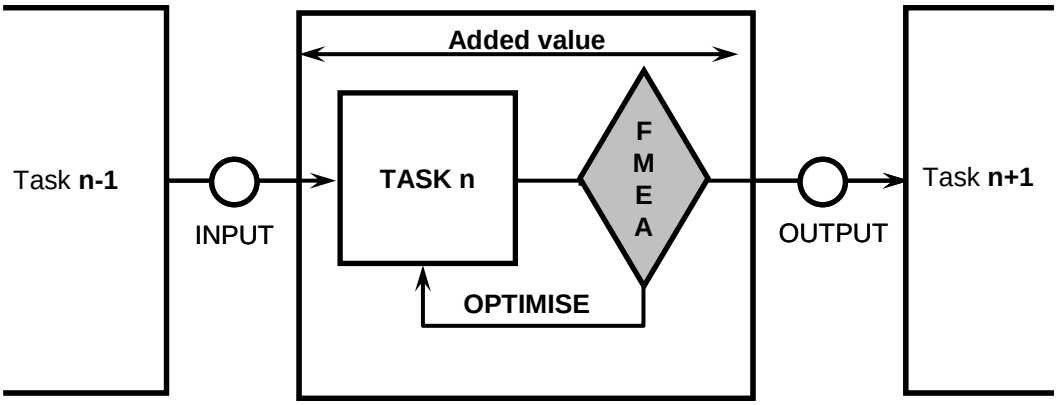

Figure 1 – Validate task by taking preventive action on potential failures

Examination of every task in this chain of activities takes place with a view to satisfying the customer / user of the product.

FMEA studies are planned in a way that they are completed before key decision-making dates in a project schedule.

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2) "IMAGINE NEGATIVE SCENARIOS", dysfunctional analysis: 3 stages (figure 2)

Stage 1:

A product, process or service is always designed with a positive outcome in mind. When we speak about a service to be offered to the customer / user of the Product, we speak in terms of customer satisfaction. This service is then broken down into technical solutions, components, characteristics, etc. with the aim of making everything run so that the customer / user of the product is satisfied.

Stage 2:

The FMEA procedure will be based on the critical or negative aspect: "what could possibly happen to make this service unavailable and to what extent would the Customer / user of the product be affected by it?" Therefore, here, we list the potential failures that make this service unavailable.

Stage 3:

The FMEA procedure is definitely not just a critical review of the product or process design. With this procedure, corrective solutions can be found that are used to **optimise the future design of the product and its industrialisation**.

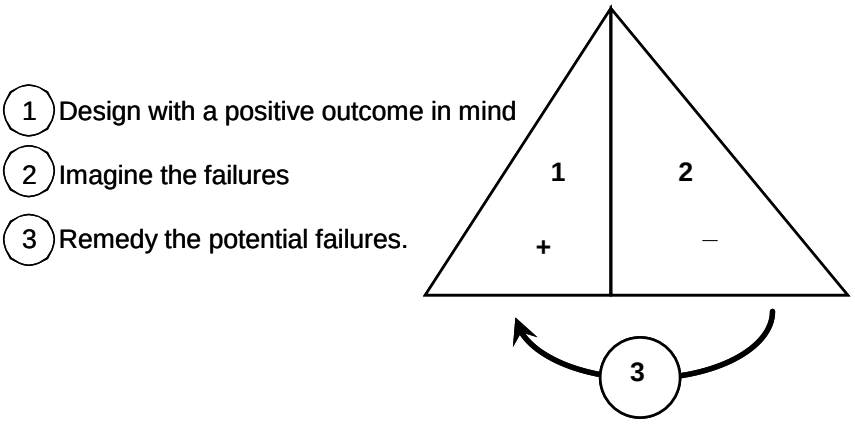

Figure 2

3) EFFECTIVENESS OF THE FMEA IS BASED ON 3 QUALITIES: COMPETENCE, CREATIVITY AND RIGOUR

Within a multidisciplinary group, the **competence** of the participants is used to explore the subject being studied. Each member shares his/her experience and knowledge.

The **creativity** of the members of the group is put to the test when they have to go beyond their experience and have to imagine potential failures and those that are the less probable.

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Rigour is developed by implementing the entire FMEA method in a systematic manner. The FMEA method can be divided into 7 stages (see figure 3 below).

STAGES	MEANING – SCOPE	WHO
1 INITIALISE	Formalise the subject on a summary sheet that is made up of: • choice of subject and its limits • group composition • planning • follow-up of results	• Coordinator • Requestor • Decision-maker • Engineering department (RENAULT, Partner or Supplier)
2 ANALYSE	• Present the FMEA and functional analysis of the requirement and of the Technical Solution • Imagine potential failures and their consequences for the customer-user of the product by using the analysis sheet	• Multi-disciplinary group and coordinator
3 EVALUATE DECIDE **REPORT**	• Evaluate and classify potential failures into a hierarchy based on predetermined Rating • Decide to start plans of action for RPN that exceed thresholds	• Multi-disciplinary group and coordinator
4 LOOK FOR CORRECTIVE SOLUTIONS	Look for corrective solutions. This action is undertaken outside the FMEA group. However, it can be outlined during FMEA meetings.	• Product and/or Process designer
5 FOLLOW UP **REPORT**	Analyse and evaluate the corrective solutions that are being studied until the RPN are lower than the thresholds.	• Multi-disciplinary group and coordinator
6 APPLY	Apply the approved corrective solutions selected. This action is undertaken outside the FMEA group at the Engineering Dept; applied to industrialisa-tion methods and during manufacturing.	• Action leader (GFE (part engineers), designer, etc.) internal or external to RENAULT
7 CHECK KNOWLEDGE-BUILDING **REPORT**	Check the effectiveness of measures taken End FMEA (signature) Build knowledge based on experience by archiving the study in a database	• Coordinator • Multi-disciplinary group • Requestor, decision-maker • Coordinator/Requestor

Figure 3

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4. DETAILED PROCEDURE FOR THE PRODUCT OR PROCESS FMEA

Implementing an FMEA consists in methodically rolling out the procedure in 7 stages. The aim of methodically applying the FMEA is to make a list of potential failures which tends toward exhaustiveness and resolving them on a preventive basis. Each stage is described below.

4.1. STAGE 1: INITIALISE

The aim of initialisation is to "present a problem" and manage it in time. This first step is fundamental for the study to be a successful one. It materialises by way of documenting the initialization sheet (appendix 1).

4.1.1. Select subjects to be dealt with

Systematizing the FMEA will only serve to validate that which is already controlled. It would lead to unnecessary added expenditure for the Company.

On the other hand, the FMEA must be used wherever there are anticipated or measured risks, innovations, incomplete information or significant issues for the Company.

The project manager (vehicle, system, unit, industrial project, production facilities, etc.) and the Quality manager draw up a **list of subjects at risk** as soon as general principles are approved at the design and industrialisation stages.

INDUSTRIAL PRODUCT OR PROCESS SET			
Known and reliable industrial product or process	Industrial product or process renewed with **recurring problem**	Industrial product or process where **risks are anticipated**	**Innovation** of the industrial product or process, **new Supplier**, regulated function/safety
Feedback	Failure or cause tree	**FMEA**	Tools and Reliability Procedure

Figure 4 – Principle on which subjects are chosen

4.1.2. Organising the FMEA study

It is obligatory for the requestor of the study (project or task "n" manager) who is internal or external to RENAULT and the FMEA coordinator to document the “initialization sheet”, which serves the purpose of a contract between the requestor and the group. It is made up of the following information:

- FMEA type (product or process)
- Vehicle, unit, scope of the study and key dates: date of request of the FMEA, VPC/PC/Pre-Contract, CO/Contract, TGA/RO/IOD, ABVC/AMS-lot or ABPT/AMEI.

  Important point:

  Design FMEA: risks must be cleared (stage 7) for TGA/RO/IOD.

  Process FMEA: the action plan must be drawn up for TGA/RO/IOD and risks must be cleared for ABVC/AMS-lot or ABPT/AMEI.
- The objective (E ‰, K ‰, S1 ‰, S3 ‰, PIMOL, durability, objective of rate of compliance etc.) and the reason for the study (see appendix 1).
- The name of the requestor (subject manager) and decision-maker (manager of important choices and decisions).
- The limits of the subject (references related to the product, process plan, etc.).
- Composition of the group (5 to 6 people):
  - 1 coordinator proficient in FMEA

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. Participants (Engineering Department, Production Engineering, Manufacturing, etc.) who are proficient in and are managers of the subject (specify the function in the initialization sheet).

- Planning the FMEA over 7 stages according to project milestone dates.
- Resolution of potential risks can be seen in the report.
- Location where the FMEA in progress and the implemented FMEA are archived.
- Study progress:
  . as per the schedule,
  . as per clearance of potential risks (stage 5)
  . as per verification of results (stage 7)

It is the responsibility of the requestor to manage the FMEA. The end of forecast studies (clearing of risks) is certified by a signature on the initialization sheet by the requester or by the decision maker.

4.2. STAGE 2: ANALYSIS

Analysis consists in studying risks that can be imagined and that are known from experience. This search is always structured on the basis of functional analysis.

4.2.1. Preparing the FMEA

The aim of this first phase is to ensure that all the members of the work group have the same level of knowledge on the subject and to get specialists to speak about the same scope.

This phase also serves the purpose of ensuring that the participants are familiar with the FMEA procedure.

Functional analysis includes (see Appendix 2):

- **the functional analysis of the requirement**, which lists, characterises and develops all service functions of the system to be studied as well as the phases in the life cycle. **Functional analysis of the requirement** (stated in the functional specifications) **is a prerequisite to the FMEA**. The FMEA group takes from it the elements necessary for the FMEA study (main and constraints functions as well as related value criteria that fall within the scope of the study). These elements are attached to the FMEA file.
- **the technical functional analysis**, which explains how the solution is supposed to operate, its organisation, during all the phases of the life cycle that are studied. This analysis is presented in the form of flow charts. Qualified and quantified operating conditions are listed. The flow chart is a result of the design phase or is created by the FMEA group.
- **the description of the industrial process** that is planned for mass producing the product. The industrial process is defined in the production and inspection process plan. Technical sheets and the description of the operating mode provide input data. This data can be seen on a process diagram (or manufacturing diagram or synoptic). Input and output conditions (Customer – Supplier relationship) are drawn up.

Using functional analysis is mandatory so that the members of the group have the same vision of the subject and can make a comprehensive list of all the potential failure causes, modes and effects that are possible as well as Design Controls.

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4.2.2. Identifying potential failures

The aim of the second phase is to highlight **all** potential failures of the studied system that can be imagined or that are known. Failures have the following qualitative and quantitative criteria:

Potential failure mode:

The potential failure mode is the manner in which the system will stop running or will run unusually.

To understand a potential failure mode ask yourself the following questions:

What are the potential failure modes that …

- no longer make the operation possible?
- no longer make the operation stop?
- disrupt the operation?
- make the operation unpredictable?
- make the operation …?

Deal with potential failure modes that may appear at 0 km **and also those that are likely to appear over time**: **reliability of the product** (consider the entire product life cycle).

Does it occur…

- totally,
- partially,
- at 0 km,
- over time (e.g.: after 3 months, 7 years, 100 000 kms, above 25 000 cycles, etc.)
- …

Identify and quantify potential failure modes of a component or a set of components on the basis of the 5 senses:

- touch: e.g.: hardness, seizure
- sight: e.g.: rough break/caused by fatigue, alteration in aspect
- hearing: e.g.: noise
- smell: e.g.: burnt smell, plastic smell
- taste: e.g.: chemical product taste

Remark:

The potential failure mode is expressed in physical terms:

- The potential failure mode arises from operating conditions (see flow chart).

- Wherever possible, quantified criteria will complement information on the potential failure mode (e.g.: noise heard within the first few kilometres (fault occurs at "0 km"), progressive breaking of the shaft under 50 daN, 100 000 kms onwards, bearing seizure 150 000 kms onwards, etc. (fault occurs "over time").

To complement and add to potential failure modes, include failures proven on the replaced product in the analysis.

Potential cause of failure:

The potential cause of failure is the initial anomaly that is likely to lead to the potential failure mode. The potential cause of failure is always an unsuitable or non-compliant characteristic of the product:

- as regards the Design FMEA, characteristics of the product that are estimated, unsuitable, missing, superfluous, etc. must be considered.

- as regards the Process FMEA, characteristics of the product that are manufactured and are estimated to be non-compliant must be considered.

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When the potential causes of failure are being looked for, it is essential to describe and formalise the entire causal chain in the analysis grid. This is the potential failure scenario.

Potential failure MODE	CAUSE of failure
Manner in which the system stops running (totally, partially, …)	· initiale cause - causal chain - … - …

+Remarks:

- Design FMEA:

  - to imagine the potential modes and causes of failure, systematically explore the entire body, contacts and the design and service flows of the flow chart.
  - for a given technical function, imagine, for example, 5 to 7 potential failure modes.
  - for a potential failure mode, imagine, for example, 5 to 7 potential causes.
  - for every potential failure mode, search for the **characteristic(s) of the product** at the source of the potential failure (explore: material, geometry, dimensioning, thermal treatment, appearance, assembly conditions, etc.); always go back to root causes; prohibit causes related to “incorrect design, incorrect wiring route, etc.”.

- Process FMEA:

  - It is necessary to imagine that the product-cause (product characteristic to be manufactured that is non-compliant or altered by the process even though it was compliant at the time of input) may lead to a potential failure mode. Then, the causal chain relative to process-causes must be searched for: "what are the process parameters that are likely to lead to a product-cause"
  - For a product-cause, imagine, for example, 5 to 7 process-causes.

Indication of the potential cause of failure in the grid:

In the analysis grid, the potential cause of failure must be given in the following manner:

Design FMEA: "product-causes" are given in the "cause” column

- **Example:**

Effect	**Mode**	**Cause**	**Prevention**	**Detection**	**S**	**O**	**D**	**RPN**
Effect E1	Mode A	Product-cause A1	Prevention PA1	Detection DA1	S1	OA1	DA1	RPN1
		Product-cause A2	Prevention PA2	Detection DA2	S1	OA2	DA1	RPN2
		…			S1	OA…	DA…	RPN3
		Product-cause An	Prevention Pan	Detection DAn	S1	OAn...	DAn	RPN4
Effect E2	Mode B	Product-cause B1	Prevention PB1	Detection DB1	S2	OB1	DB1	RPN5
		Product-cause B2	Prevention PB2	Detection DB2	S2	OB2	DB2	RPN6
		…			S2	…	…	RPNn

Process FMEA: indicate "product-causes" (non-compliance during manufacturing) in the "cause" column, then, under each product-cause indicate “process-causes”, which lead to the appearance of the non-compliance. Product and process-causes are given in the “cause” column.

- **Example:**

Effect	Mode	Cause	Prevention	Detection	S	O	D	RPN
Effect E1	Mode A	• Product-cause A1	Prevention PA1	Detection DA1	S1	OA1	DA1	RPN1
		- Process-cause A1p1						
		- Process-cause A1p2						
		- …						
		- Process-cause A1pn						
		• Product-cause A2	Prevention PA2	Detection DA2	S1	OA2	DA2	RPN2
		- Process-cause A2p1						
		- …						
		• Product-cause An	Prevention PAn	Detection DAn	S1	OAn	DAn	RPNn
		- …						

Potential effect of failure:

In the Design and Process FMEA:

The effect of a potential failure is defined by 2 parameters:

- on the one hand, it is related to a **disrupted service function**,
- on the other, it states **the potential consequences for the final user of the vehicle**.

Note: If necessary, this standard is also used to indicate the effect that product non-compliance can have on manufacturing/assembly. In this case, the effect on the customer / user of the product must also be indicated.

It is the potential failure mode that leads to the disruption of a service function.

Potential EFFECT of failure		**Potential failure**
Potential consequence for the customer / user of the product	Disturbed service function	**Mode**
Dissatisfaction of the customer / user of the product	*Service function totally, partially, etc. disturbed*	*Manner in which the system/the component stops operating (immediately, over time, etc.)*

Remarks:

- The wording of the disrupted service functions is chosen on the basis of the potential failure mode and the functional analysis of the requirement.
- In order to be as thorough as possible, imagine:
  - . no service function,
  - . loss of service function,
  - . unpredictable triggering of the service function,
  - . degraded service function (over time, at 0 km, etc.),
  - . service function that is misinterpreted, etc.
- For the Design and Process FMEA, the consequences for the customer / user of the product are stated on the basis of the table that appears in Appendix 4.

In the flow FMEA, the manufacturer and central production departments are the Customers: DLI (see appendix 5)

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Design Control:

Design Control is an action planned, in progress or already implemented so as to prevent the potential failure cause (and/or potential failure mode) from appearing and affecting task N+1.

Design Control is an activity that is carried out by the designer of the product or process to check whether the design of the product or manufacturing process is in compliance in relation to the imagined potential failure scenarios.

*** Remark:**

The later non-compliance is detected, the more considerable business consequences are. Therefore, for business reasons and as the responsibility of each department, in a Design and Process FMEA, all potential non-compliance must be detected before going to the next stage in the design/industrialisation/manufacturing process. This next stage is called “Task N+1” in the rest of the document.

Task N+1 is the stage which comes immediately after (downstream from) the analysed activity. So, it does not yet relate to the customer / user of the product (see figure 1).

Examples:

- In the Design FMEA, Task N+1 is an activity that falls under the Industrialisation Methods Dept (manufacturing process engineering dept).
- In the Process FMEA, Task N+1 is an activity that falls under the Manufacturing sector, downstream from the studied process.

"Do not generate a defect, do not accept a defect, do not deliver a defect."

Potential failure MODE	Potential CAUSE of failure	Design Control
Manner in which the system has stopped operating (totally, over time, etc.)	• Initial cause: - causal chain - ..... -	**That which is planned, in progress or already implemented** to prevent the initial potential cause of failure (and/or the potential failure mode) from appearing or affecting task n+1.

There are 2 types of Design Control:

- **Actions related to Prevention that aim to prevent the potential cause or failure mode from appearing; give priority to theoretical Prevention actions.** From among the different **Design Controls**, certain actions can also be qualified as Prevention actions when they are implemented with a view to optimising a technical solution so that the potential cause or failure mode does not appear. They work towards lowering the **Occurrence index O**:

  Example for the Design FMEA:

  . analytical study (calculations, chain of dimensions, etc.)

  . simulation program at limit states, digital simulation, digital mock-up design review (DMDR), etc.

  . test campaigns (design of experiments)

  . correlation of calculations/testing

  . "Robust Engineering" study

  . compliance with proven design rules, etc.

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Example for the Process FMEA:

. digital simulation, digital mock-up design review (DMDR), etc.

. transformation of functional tolerance into manufacturing tolerance based on chains of dimensions

. operator training

. increasing the capability of the process

. setting up an anti-error device (poka yoke), etc.

- **Actions related to Detection that aim to detect the potential cause or failure mode that is supposed to have occurred; give priority to theoretical Detection actions. Detection** actions aim to detect the potential cause or failure mode that is supposed to have occurred and work towards lowering the Detection index **D**.

  Example for the Design FMEA:

  . Chain of dimensions to detect interference in a theoretical manner

  . Creating a mock-up of a solution to check whether it is operating correctly

  . Functional testing at limit states

  . Bench testing, durability testing during operation at limit states, etc.

  Example for the Process FMEA:

  . Simulating the injection of a plastic part to detect areas to which the material has not spread in a theoretical manner

  . Checking the compliance of a characteristic, 100% checking or by sampling

  . Checking a process parameter

  . Due to Product Safety or regulation special characteristic, the Process must be robust in order to produce this S or R characteristic 100% compliant.
  Example:
    . Control of existence of fuel leakage hole with the specified diameter marked S.
    . A tightening torque, a temperature, a pressure, a set point (e.g. push / pull, check for presence) are process characteristics.

  . Functional testing, etc.

To find out more about the Safety and Regulation Special Characteristics, refer to the standard 00 - 10 - 040 : DEFINITION, IDENTIFICATION, OBLIGATIONS AND TRACKING OF PRODUCTS AND SAFETY OR REGULATORY CHARACTERISTICS

The safety characteristic is linked to one or more potentially Unwanted Costumer Event Potentially Safety related (UCEps), these being defined in the approved list of UCEps En_RPIFFSGPR20150009-V4_AEK-D-00051_2.0 and generally identified for a system given through the completion of a PRA.

Distinction made between Prevention actions and Detection actions in the analysis grid:

- **Prevention actions are entered in the "Prevention" column**
- **Detection actions are entered in the "Detection" column**

Remarks on Prevention and Detection actions:

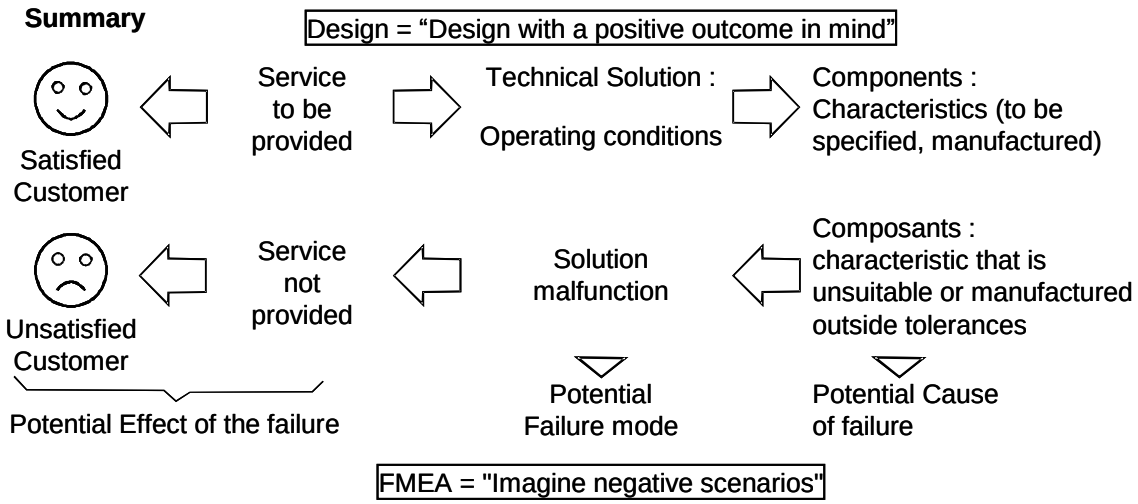
- In the "Prevention" and "Detection" columns, enter actions that are **planned, in progress or already implemented** (**those that have to be planned** will be described in the **action plan** column, if needed).
- In the Design FMEA as well as in the Process FMEA, in order to reduce costs, give priority to “theoretical” actions (calculations, chains of dimensions, simulations and digital design review, project reviews, recommendations from Technical Departments and experts, etc.) as far as possible, over “physical” actions (prototypes, bench or vehicle testing, 100% inspection, etc.).
- Document actions in the most complete manner as possible (qualify and quantify parameters).

Analysis grid:

The following can be saved by using the analysis grid:

- references of the subject, names of analysts, dates of analysis, evaluation and follow-ups.
- subject data to be processed.
- list of potential failures and Design Controls for Prevention and Detection.

Note: The grid is compiled on A3-size paper or online (see Appendix 3).



4.3. STAGE 3: EVALUATE - DECIDE

Evaluation consists in quantifying the level of risk of each potential failure in order to help the requestor, decision-maker and participants of the study to define corrective actions that have to be given priority.

Evaluation is conducted once the entire analysis (effects, modes, causes and design controls) has ended. A ranking table lays down the limits that define acceptable risks and those that are not.

4.3.1. Principle of evaluation:

Evaluation is based on the sequence of events that leads to the customer / user of the product perceiving a failure and to consequences that impact him/her (see figure 5).

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For a potential cause of failure to lead to a potential failure mode that impacts the customer / user of the product, the 3 following conditions need to be fulfilled:

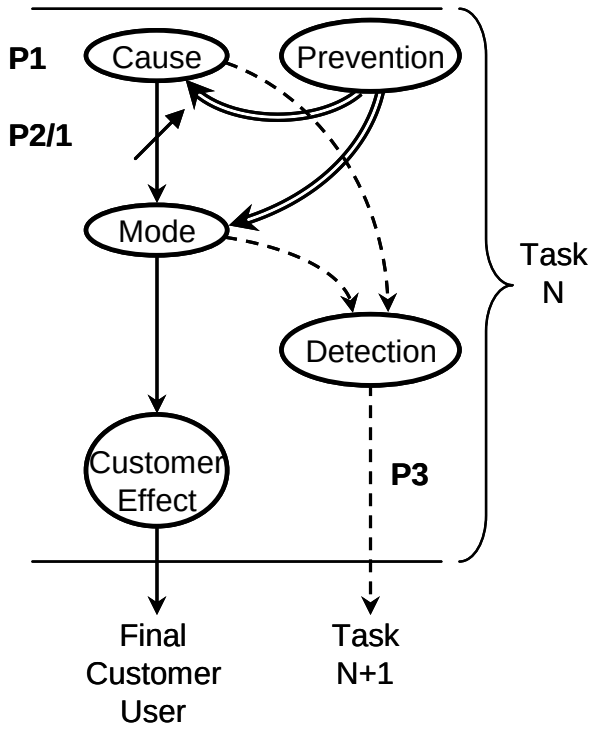

Figure 5 – Chain of events

1. The potential cause of failure is supposed to have appeared. Probability P1 may be attached to this event.

2. If the potential cause of failure has appeared then it must necessarily lead to the potential failure mode that is considered. The probability attached to this event is conditional probability P2/1.

3. The potential cause-mode pair of failure that is supposed to have appeared will have to have passed Detection for it to affect Task n+1. Non-detection probability is called P3. P3 characterises the permeability level of Detections that are planned or set up by the person in charge of task N.

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Remarks: it is necessary to consider that after task N, there is no more Detection. Due to this, as regards a Process FMEA for example, even if the potential failure can be detected because assembly by the Manufacturing sector has been made impossible at the stage that is downstream from the studied process stage (N+1 or N+2, etc.), this downstream Manufacturing sector will suffer the consequences in the form of production losses, line stoppage, vehicle blocked in plant stock before delivery, etc. because it is assumed that the parts in the manufacturing sector process concerned are compliant.

However, it is possible to have a contract drawn up between the person in charge of task N and the person in charge of the downstream task (N+1 or N+2, etc.) so that Detection is officially set up and included in the scope of the person in charge of the downstream task, who then becomes responsible for it.

If P is the probability that the customer / user of the product is affected by, it can be expressed in the following manner: P = P1 x P2/1 x P3

4.3.2. Ranking indexes

Ranking is based on 3 indexes and on predetermined ranking, namely:

- **Severity index S:** is the ranking of the consequences of each potential failure perceived by the customer / user of the product (see Appendix 4).
  In the Process FMEA, it is also possible to indicate in a timely manner the effect of product non-compliance has on manufacturing/assembly. In this case, for every potential failure that has an effect on manufacturing/assembly, the consequences for the customer / user of the product must also be indicated and both will have to be dealt with.

- **Occurrence index O:** corresponds to the probability that the product cause appears and leads to the mode. It arises from predetermined ranking and is a function of the product P1 x P2/1. When an Occurrence index is attributed, it is necessary to include these 2 probabilities (see tables in Appendix 4).

  Design FMEA: the Occurrence index should be estimated based on Prevention actions which were identified during analysis, the Occurrence ranking table given in Appendices, and on:

  - feedback in the case of a reused technical solution
  - feedback and knowledge in the case of a solution based on basic solution elements that are already known and implemented (as in the case of improvements or innovations)
  - experience gained in the case of inventions, during design, through mock-up creation, testing, design of experiments, durability testing conducted in an environment that represents mass production of the vehicle.

  Process FMEA: the Occurrence index should be estimated based on Prevention actions which were identified during analysis, the Occurrence ranking table given in Appendices, and on:

  - probability P1 for appearance of the cause defined by:
    - process dispersion related to a characteristic when the manufacturing process has already been implemented in a similar context,
    - or estimating dispersion when the manufacturing process is new.
    - P1 is estimated for each product cause-mode pair: it takes into account the impact of each process cause on the product cause.
  - and, probability P2/1 for appearance of the potential failure mode (same for Design FMEA)

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- **Detection index D:**

  - it corresponds to the probability that the product cause (and/or mode) that is supposed to have appeared affects task n+1:

    - In the Design FMEA, it is inferred from the risk percentage that is estimated during product design that a product characteristic, which may lead to a potential failure, has not been identified and intercepted before affecting Product/Process design task N+1.

    - In the Process FMEA, it is inferred from the risk percentage that is estimated during manufacturing process design that a manufactured product characteristic that is not in compliance with the drawing, has not been identified and intercepted before delivery to the manufacturing sector following the studied process.
      D is estimated for every product cause-mode pair and takes into account the impact of each process Detection on process causes (if such Detection exists).

  - Detection is estimated using ranking (given in Appendix 4) and corresponds to P3.

  Design FMEA: the Detection index should be estimated based on Detection actions which were identified during analysis, the Detection ranking table given in Appendices; and on:

  - Technical department rules, feedback, recommendations from experts, etc.

  - application of robust engineering, dimensioning calculations, functional chains of dimensions, integration of production dispersion and off-centring (e.g.: Cam/Cap and Cpk, etc. that have been forecast so that the distribution of tolerances is optimised, project review, digital simulation, digital mock-up creation, physical prototypes, etc.

  - results of a testing plan, validation in compliance with Technical department's recommendations (validation plan), compliance with regulations and safety norms, etc.

  - testing in compliance with the usage of the product by customers

  - correlated results from calculation/bench testing/actual operation, etc.

  Process FMEA: the Detection index should be estimated based on Detection actions which were identified during analysis, the Detection ranking table given in Appendices; and on:

  - transformation of functional tolerance into manufacturing tolerance, capability studies, 100% or frequency-related inspection, Statistical Process Control, inability to assemble, impossible to continue production process, unit testing or end of the line, etc.

  - Technical department rules, feedback, recommendations from experts, etc.

  - Note: detection encompasses a larger area than inspection (a 100% inspection may prove to be entirely ineffective in terms of detection. E.g.: 100% functional inspection to detect a failure mode that appears over time, after 100 000 kms/5 years).

Ranking tables for S, O and D:

- Design and Process FMEA: ranking varies from 1 to 10; see appendix 4.

Note 1: during evaluation, it is very important to define ranking values related to Severity, Occurrence and Detection up to the most precise value; however, in case doubt exists between the 2 values (e.g.: doubt between S = 7 or S = 8), the higher value must be considered so as to prioritise the protection of the customer / user of the product.

Note 2: a common error is to assume that because a phenomenon is not going to occur, the Severity to be associated with it must be low: **do not confuse failure criticality indicated by the Risk Priority Number (RPN) with the Severity level indicated by S.**

Note 3: as regards evaluating P2/1, this probability is always of interest: it contributes to the evaluation of the Occurrence index O, up to the most precise value. It may so happen that a

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probability cannot be associated with this event (P2/1). In this case and even though it may be penalising, it is possible to consider this value equal to 1, that is, the cause always leads to the mode.

Note 4: The robustness of product design is a quality of product definition that is applied to the technical drawing. This makes it more or less sensitive to non-compliance that appears during manufacturing. If the characteristic is within the tolerance, no failure mode will appear.

But, by way of example, one may say that:

- for a so-called robust solution, even if the characteristic exceeds the tolerance, no failure mode will appear before deviating for example from 1x, 2x, 3x, Nx, etc. the tolerance. In this case, it can be said that the product definition that is applied to the drawing is robust since Non-Compliance does not lead to a failure mode. In this case, the opportunity to increase the tolerance value might have to be examined. In FMEA (product and process alike), in this case, if the potential cause of failure appears but does not lead to the failure mode: P2/1 is considered to tend towards 0.

- conversely, if we imagine that as soon as a characteristic exits the tolerance, the failure mode appears, this would mean that the tolerance value is at the maximum. This could also mean that the product definition applied to the drawing is very sensitive to Non-Compliance. In the FMEA (product and process alike), in this case, if the potential cause of failure appears, it immediately leads to the failure mode: P2/1 is considered to tend towards 1.

Note 5: certain design controls may have an incident that simultaneously occurs on the Occurrence index and the Detection index. Take these cases into consideration when evaluating O and D. E.g.:

- in the Design FMEA, a chain of dimensions may lower the Occurrence index so as to have dimensions that do not cause interference between the parts and the mechanism. It is also made up of theoretical Detection that is efficient in finding the interference risk.

- in the Process FMEA, simulation of plastic part injection to ensure no areas on the injected part are free from materials will enable similar action to be taken on the Occurrence of these areas without material appearing and their Detection.

4.3.3. Calculating criticality

-By Risk Priority Number « RPN »:

Every time "potential effect of failure, potential failure mode, potential cause of failure and Detection" are associated, multiply the 3 ranking values: Severity, Occurrence and Detection.

The result is the **R**isk **P**riority **N**umber that is also called Criticality.

RPN = S x O x D.

Evaluation is assessed by the members of the group based on databases and by default, on their knowledge and experience.

Probabilities P1, P2/1, P3, ranking values and RPN are saved and kept on analysis sheets.

- By Action Priority « AP» (AIAG VDA FMEA Handbook) :

The criteria S, O, D and criticality "AP" of the international manual "AIAG / VDA FMEA HandBook" can be used for the prioritization of risks (instead of RPN).

4.3.4. Classification

To obtain an overall image of the level of risk of the studied system, the RPNs are represented in the form of a bar graph.

This representation makes it easier to follow up improvements that will be made later.

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Design and Process FMEA

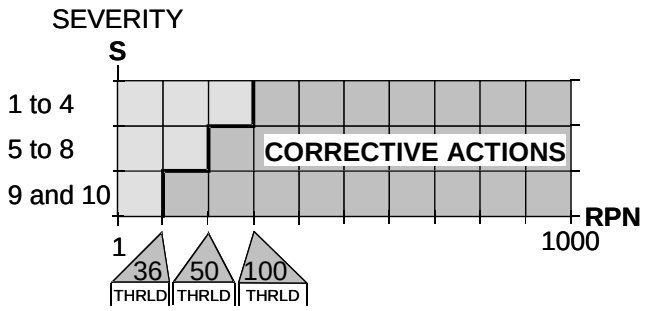

Figure 6

4.3.5. Deciding corrective actions on a priority basis

The purpose of this decision is to implement everything that would make it possible to:

- meet objectives related to compliance, quality, reliability, durability (refer to the FMEA initialization sheet).
- optimise the Cost of Quality.

Decision in terms of thresholds:

Corrective actions are undertaken for all RPNs that exceed the threshold set by this standard.

TYPE OF FMEA	CORRECTIVE ACTION FOR:
Product and process	RPN > 36 when S = 9 and 10 RPN > 50 when S = 5 to 8 RPN > 100 when S = 1 to 4

A lower RPN can be obtained by acting on the indexes: O or D or both.

- Acting on O as priority in such a way that the potential failure does not occur (Prevention is better than cure).
- Acting on D, as a complement, if needed: the potential failure that is supposed to have appeared, what has been planned to identify and intercept it before it affects task N+1.

Index and probability		Design and Process FMEA
S		**It is rare to be able to directly act on consequences for the customer / user of the product unless the technical solution is modified in such a way that the effect produced by the failure mode on the customer is minimal.** Examples: - before the technical solution is modified, it is impossible to close the sunroof because part P is broken. - after the technical solution is modified, breakage of the same part P makes it impossible to open the sunroof, not close it.

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Index and probability		DESIGN FMEA
O	P1 P2/1 P1 & P2/1	Act on Occurrence as a priority, that is: - Reduce potential causes of failure. - Act on product design, optimise the functional characteristics of the product (dimensions, geometry, material, thermal treatment, aspect, protection, etc.) Increase robustness of the product design: the solution should not be very sensitive to variations in tolerances or add redundancy, etc. Combine both actions
D	P3	**Secondly, act on detection only if needed.** Set up or improve Detection (calculations, chain of dimensions accompanied by a review of dispersions, digital simulation that is correlated with physical tests, design of experiments, application of Technical Dept rules, constraint/resistance diagram, recommendation from an expert, producing prototypes, representative durability tests: vehicle, unit or component, in the laboratory, on the bench or in the vehicle, etc.). Note: in complex scenarios, set up progression that goes from the theoretical to the physical as regards Detection. Vehicle or unit testing is only planned to demonstrate compliance. Example 1 for progressive Detection: 1. dimensioning calculations 2. chains of dimensions 3. drawing review 4. digital simulation 5. mock-up creation 6. physical prototype that is representative of mass production or not 7. unit testing, destructive testing, etc. 8. vehicle testing Example 2 for progressive Detection: 1. digital simulation 2. recommendations from technical departments and experts 3. physical prototype NOTE: In complex and exceptional scenarios, when the result can only be obtained with components made during a process that is representative of mass production, the result will be planned in accordance with the project milestone.

Index and probability		PROCESS FMEA
O	P1 P2/1 P1 & P2/1	**ACT ON OCCURRENCE AS A PRIORITY, THAT IS:** Reduce potential causes of failure: 1. Act on process causes by controlling parameters related to production facilities and processes (tooling design, operating mode, cutting speed, etc.) or by optimising the manufacturing process plan. 2. If this is not sufficient, act on Product causes (characteristics of the product) by optimising the functional tolerancing and the manufacturing tolerancing. Increase robustness of the product design: the solution should not be very sensitive to production dispersions or add redundancy, etc. Combine both actions
D	P3	**SECONDLY, ACT ON DETECTION ONLY IF NEEDED.** Improve or set up Detection (digital simulation, design of experiments, capability of production facilities, error-proofing devices, intensified monitoring plan, 100% sorting, functional testing, random auditing of processes, etc.). In complex or critical scenarios for task N+1 or for the customer / user of the product, set up mass production Detection to lower P3.

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Optimising the cost of quality:

The cost objective is brought up at this stage in the FMEA, in 2 respects:

- Reduce costs due to over-quality.
- If a choice has to be made between 2 solutions having the same predicted level of quality, opt for the less costly one while taking into consideration investment and operation.

Solution	S	O	D	RPN
A	1	1 P2/1 = 1 ; P1 = 0	1 P3 = 0	1
B	7	10 P2/1 = 1 ; P1 = 0.3	1 P3 = 0	70
C	7	1 P2/1 = 1 ; P1 = 0	10 P3 = 1	70

Remarks:

Solution A: Isn't the solution too robust? Why plan 100% effective Detection for a failure that has very low probability and has no impact on the customer / user of the product?

Solution B: manufacturing the product with non-compliance estimated at 30% and 100% efficient Detection (no potential non-compliance will affect task n+1).

Solution C: manufacturing the product with 100% compliance and no planned Detection.

With regard to predicted quality where there is equivalence between B and C, only a business study that considers operational investment and operating cost must make it possible to select the most economic solution without forgetting the risk to the customer / user of the product.

For all corrective actions that have been decided upon, an action plan needs to be formalised on the analysis sheet:

- Write down measures that are recommended to reduce the RPN (act on P_1 and/or $P_{2/1}$ and/or P_3).
- Write down the name of the person in charge of the action (his/her department) and the date on which the studied corrective solution is presented.

Action plan	
Recommended measure	Manager Technical dept Target Completion Date

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Examples:

DESIGN FMEA:

RPN	Recommended measure	Manager Technical dept Target Completion Date
RPN > threshold	• Reduce P1: e.g.: ○ add geometric tolerances in boring ○ add a cylindricity tolerance of 0.01… • and reduce P2/1: e.g.: ○ study a solution with roller bearings instead of a solution with ball bearings… • and reduce P3: e.g.: ○ calculate constraint on the bearing ○ conduct intensified bench testing for durability: 200 000 cycles subjected to stress (10 DaN) at 80°…	M. Durant Dept … Date on which presentation is made: ….

PROCESS FMEA:

RPN	Recommended measure	Manager Technical dept Target Completion Date
RPN > threshold	• reduce P1: e.g.: ○ define min and max parameters for adjusting welding progress ○ optimise tolerance of manufactured dimensions: ±0.15 instead of ±.0.2 • and reduce P2/1: e.g.: ○ study the possibility of placing a 3rd support in position to limit part sagging • and reduce P3: e.g.: ○ inspect (once every hour) welding torch progress ○ save and analyse parameters ○ conduct a destructive test (once per shift), analyse and save results, etc.	M. Durand Dept … Date on which corrective solutions were presented: …

Important point: The RPN must not be re-evaluated at this stage but at stage 5.

The FMEA group can estimate the potential gain in relation to the RPN threshold to be reached by considering that the action plan will be implemented with positive results. **However, at this stage in FMEA progress, do not place expected results in the "Measure taken" column: the action that is actually taken is not yet known.**

4.4. STAGE 4: SEARCHING FOR SOLUTIONS

Looking for corrective solutions is generally carried out outside the FMEA group, especially when major modifications must be made to the product and process. However, it may be outlined within the FMEA group.

From the action plan drawn up at the previous stage, the designated manager makes sure that resources are used to construct and formalise corrective solutions; examples:

- a new technical solution is tracked
- functional definition of the product is optimised on the drawing,
- chains of dimensions are produced and are compliant,
- results of simulations are in accordance with expectations,
- the manufacturing process plan is modified,
- parameters of facilities are defined based on quantified data,
- the monitoring plan is formalised…

IMPORTANT POINT:

Corrective solutions must be **constructed and formalised** in order to be presented to the FMEA group at stage 5.

4.5. STAGE 5: FOLLOW UP

Follow up consists in analysing and evaluating the proposed corrective solutions (figure 7), as given previously (§ 4.2. and 4.3.).

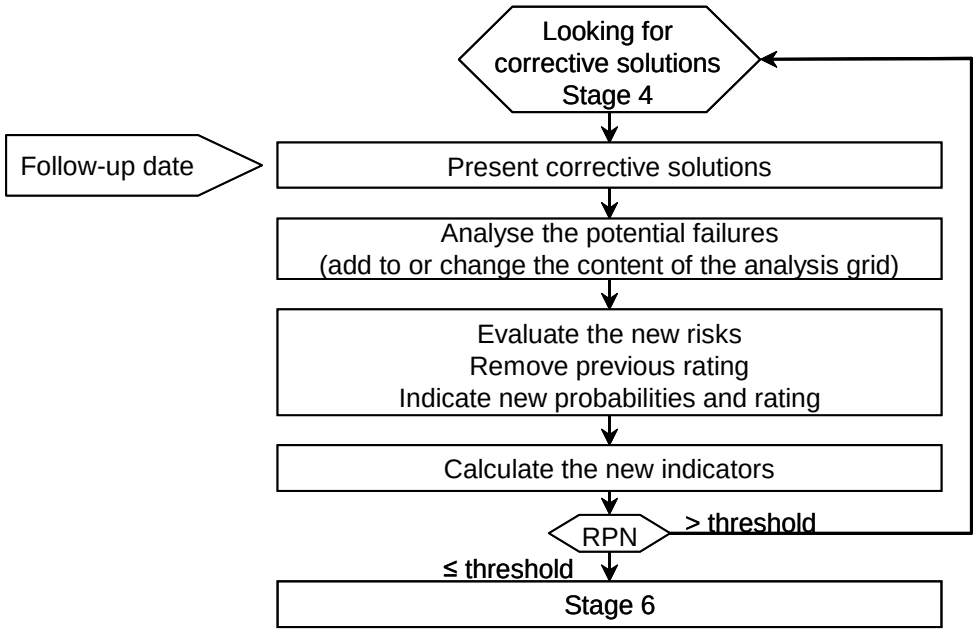

Figure 7 - Procedure

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This stage in which potential failures are corrected continues until all RPNs at risk reach a level that is lower than the set thresholds (figure 8).

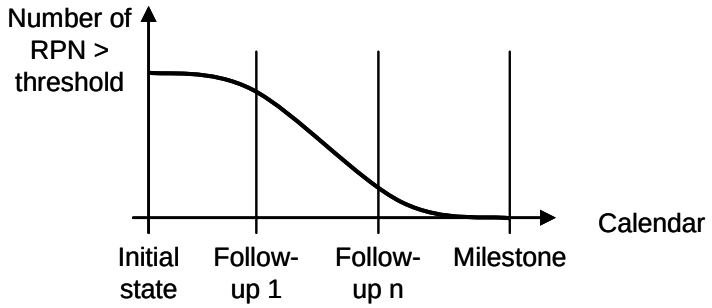


Figure 8 – Resolution curve

IMPORTANT NOTE:

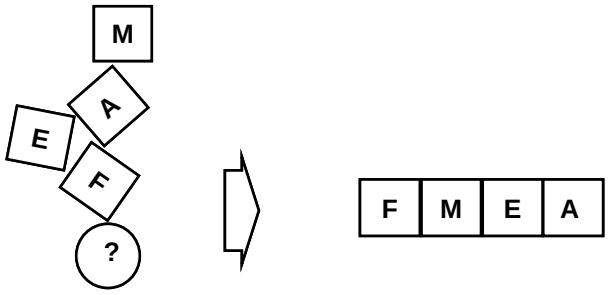
Following up corrective actions is an aspect of paramount importance in the effective clearing of identified risks.
The requestor of the study is responsible for ensuring regular FMEA progress and the clearing of risks.

4.6. STAGE 6: APPLY

Corrective solutions that were validated during the follow-up are applied.

Examples:

- specification of the new technical solution
- application for modifying the definition
- validation of calculation, simulation results
- validation of a positive testing result
- modification of a manufacturing process plan, procedure, tooling, etc.
- specification of a provisional monitoring plan
- modification of operating modes
- ...



Prior to …… and after corrections

Highlighting potential failures

Implementing the measures taken and obtaining a product in compliance with customer expectations

Figure 9

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4.7. STAGE 7: CHECK – BUILD KNOWLEDGE

4.7.1. Principle of checking

Checking ensures that the objectives that were set when the " initialization sheet" was drawn up are met.

It is carried out on 2 occasions, for Purchased Parts and Internally Worked Parts:

1- **During the design phase,** at every quality review at milestones and especially:

- for the Design FMEA: at TGA/RO/IOD ; risks must be cleared (stage 7) for this milestone.
- for the Process FMEA:
  - at the ABVC/AMS-lot (Vehicle Engineering perimeter)
  - at the ABPT/AMEI (Mechanical Engineering perimeter)

  in order to ensure that the corrective actions are in place and effective ; risks must be cleared (stage 7) for these milestones.

2- **During mass production phase,** to estimate the efficiency of the FMEA study in relation to failures proven during manufacturing and at the customer-site.

4.7.1.1. Verification on the basis of a diagnosis grid at the time of every quality review at milestones

Based on the study, members of the FMEA group:

- check, in practical terms, whether the corrective solutions are in place,
- confirm with measurements, every time it is possible, hypotheses accepted during provisional evaluation.

For example: implement capability of the facility to determine the actual P1 level, check the accuracy of the chain of dimensions and related calculations, check flow time, in-process materials, etc.

Note: The group takes ownership of the FMEA. Every member must comply with FMEA commitments and modifications made to the FMEA file must be a result of a concerted effort.

Results of the check are recorded in a summary note that is attached to the FMEA file. It is a part of the Quality Assurance file.

When one project milestone goes to the next one, a diagnosis (see appendix 6) is strongly recommended so that FMEA compliance can be attested in relation to the present standard (or in relation to a standard that is deemed to have equivalence: at this stage, formalise deviations with regard to the criteria used for clearing risks and criteria in the RENAULT standard).

The requestor of the study, or the decision-maker if needed, decides on the end of the FMEA and clearing risks by signing the initialization sheet after checking the results presented by the group.

At the milestone where risks should be cleared (stage 7), 2 cases appear:

- **Case 1:** if all RPN are lower than the thresholds: the requestor signs the initialization sheet so as to close the forecast study after having checked whether all potential risks have been cleared (no RPN exceeds threshold); evidence is based on **facts**.
- **Case 2:** if certain RPN exceed thresholds: the decision-maker (project team) must be informed about the potential risks that have not been cleared or that cannot be cleared by the FMEA team contingent on proof of confidence provided by the FMEA team (level to which a risk has been cleared, proof presented) and project issues. This will help him/her make recommendations as regards continuing the project:

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- if the level of risk is considered to be acceptable by the decision-maker or the project team, the decision-maker signs the initialization sheet to certify that the risks have been taken into account. An additional action plan is decided upon if necessary.

- if the decision-maker or the project team considers the level of risk to be unacceptable, the decision-maker does not sign the initialization sheet. An additional action plan is decided upon or the project is stopped.

4.7.1.2. Checking during mass production phase

- During manufacturing: the FMEA file is included as a part of the process audit repository.

- On the one hand, the sustainability of corrective actions can be checked and on the other hand, effectiveness of the FMEA study, by comparing the deviation between potential failures and failures proven on the process.

4.7.2. Building knowledge based on experience

The contents of an FMEA are a written record for the Company:

- that can be used for future projects (as is or to provide additional information),

- that can be used during process audits in the operation phase,

- that can be used when the product or process undergoes changes (phase II, increase in the number of production centres, etc.),

The FMEA standard is reviewed on a regular basis in order to include changes made to Company requirements and strategy.

The FMEA study must be archived in a place where it can be accessed for future use (examples: intranet, BMIR, IG database, E-Room, etc.). It constitutes one of the elements that contribute towards proving that the designed product is reliable.

In addition to this, a FMEA file must be compiled and archived in paper or electronic format and will include the following:

FMEA:	product	process
Summary sheet.	X	X
Data from the studied subject and modifications:		
- specifications or its reference,	X	X
- drawings, diagrams or their references (modified drawings),	X	X
- manufacturing process plan or reference (the modified manufacturing process plan and monitoring plan)	X	X
Functional analysis:		
- study on functions,	X	X
- flow chart updated with modifications made	X	X
- process diagram updated with modifications made,		X
Sheets related to analyses, evaluation and corrective actions.	X	X
Summary notes.	X	X

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APPENDIX

APPENDIX 1: NORMATIVE: INITIALISATION SHEET

RENAULT	**FMEA INITIALIZATION SHEET**	Design FMEA ☐ Process FMEA ☐ Other FMEA:

Vehicle:
Unit:
Scope:

Request date:
Milestone date (TGA, ABVC/AMS-lot or ABPT/AMEI):
Production date:

Objectives of the study:

Causes of the study:

UCEps identified by the preliminary risk analysis (PRA):

REQUESTOR: M. Dept	DECISION-MAKER: M. Dept

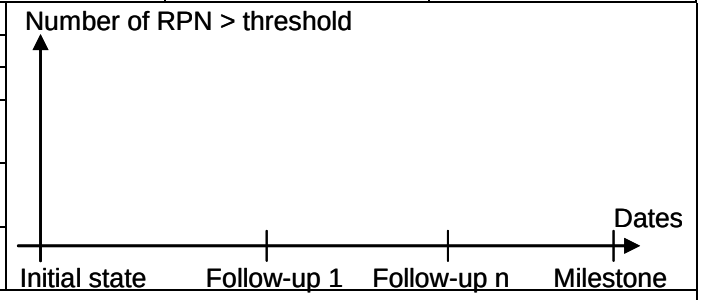
Limits of the study:

PARTICIPANTS: Permanent members MM. Dept and function	Temporary members MM. Dept and function

COORDINATOR: M. Dept

PLANNING																													
	Weeks																												
	Forecast																												
	Completed																												
	Key	Meeting: M stages 2 and 3							Follow up: F stage 5							Check: C stage 7							Milestone date: Mil						

REPORT			Initial	Follow up		
		Date:				
	RPN No.	TOTAL				
		S 9 to 10: RPN > 36				
		S 5 to 8: RPN > 50				
		S 1 to 4: RPN > 100				



Archive location:

Decision to end study:

Date:
Signature of the Requester or Decision Maker:

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APPENDIX 2: NORMATIVE: LISTING SOURCE DATA AND FUNCTIONAL ANALYSIS TO BE CARRIED OUT

INPUT DATA	TYPE OF ANALYSIS TO BE USED	FMEA	
		PRODUCT	PROCESS
Requirement or functional specifications	Functional analysis of requirements (function related to use, adaptation, assessment, constraint function and value criteria)	YES	YES
Drawings of the studied system	Technical functional analysis of the design (Flow Chart, analysis of kinematic links, Table of Technical Functional Analysis: TAFT)	YES	YES
Process plans: - manufacturing - inspection - ... with references	Process diagram (or manufacturing and inspection diagram)		YES

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APPENDIX 3: NORMATIVE: PRESENTATION OF THE ANALYSIS GRID

Design FMEA, Process FMEA, Flow FMEA

	Failure Modes, Effects and Criticality Analysis																	
	Unit … Vehicle 1			FMEA 2	Analysts : 3							Dates : 4						
	References :																	
N°	Function And/or process	Potential Failure				Design Controls		S	O	D	RPN ou AP	CSR	Action Plan					
		Mode	Cause	Effect		Prevention	Detection						Recom-mended measures	Manager. Dept Target Completion Date	Result			
		(P2/1)	(P1)	Disturbed function	Customer consequence		(P_3)								Action taken	S	O	D	RPN or AP
5	6	7	8	9		10	11	12				13	14	15	16	17		

Caption:

1. Product name
2. Specify: Design, Process or Flow FMEA
3. Names of FMEA participants and coordinator
4. Source dates, and dates on which changes were made to the document
5. Number used as a line marker for each potential cause of failure.
6. Studied functions (Design FMEA) or descriptions of manufacturing operations (Process FMEA) or flow management points (flow FMEA)
7. Potential failure mode: states the manner in which the system no longer functions.
   Also write down the probability value in this column: P 2/1.
8. Potential cause of failure relative to the product: initial anomaly that may produce the potential failure mode.
   The causal chain is the cascade of events that is of importance when the product-cause appears.
   Also write down the probability value in this column: P 1.
9. Disturbed function and consequences for the customer / user of the product or manufacturing/assembly
10. Prevention control: provisions planned in order to prevent the cause or mode from occurring
11. Detection control: planned provisions for preventing the cause (and/or mode) that may appear from affecting task n+1. Also write down probability value P3 in this column.
12. Quantify indexes related to Severity, Occurrence and Detection on the basis of ranking tables and calculate criticality level (RPN):

- S: Severity index: see Severity ranking.
- O: Occurrence index: see Occurrence ranking and see product of probabilities P_1 x $P_{2/1}$.
- D: Detection index: see Detection ranking and see probability P_3
- RPN= S x O x D
  OR
- AP: refer to the AIAG/VDA FMEA HANDBOOK

13. CSR: Proposal for S or R Logo of a Product or Process characteristic (Process FMEA only)
14. Write down the measures that are planned to reduce the RPN (act on P_1 and/or $P_{2/1}$ and/or P_3).
15. Name of the person in charge of the action and the Target Completion Date for presenting the proposed corrective solution.
16. Record of the corrective measure presented at the time of analysis and evaluation (stage 5).
17. Calculation of the new RPN or AP after the proposed corrective solution has been analysed and evaluated.

The appendix 3, FMEA analysis grid in Excel format is attached to this PDF

Appendix 4: normative: ranking tables

Design and process FMEA – severity index S

S	General defect types	Criteria to be used in Design and Process FMEA	
		Effect on the customer / user of the product	**Disturbed service function**
1	Defect tolerated or not perceived by the customer	The customer does not notice the defect.	No obvious deterioration in vehicle or equipments performance.
2		The customer tolerates the defect.	No obvious deterioration in vehicle or equipments performance.
3	Minor defect leading to product approval comfort defect	The defect, which the customer considers to be minor, leads to a defect impacting the driving comfort; the cost of repair is very low and can wait until standard service.	The failure leads to a very minor deterioration in vehicle or equipments performance.
4		The defect, which the customer considers to be minor, leads to a defect impacting the driving comfort; the cost of repair is low and can wait until standard service.	The failure leads to minor deterioration in vehicle or equipments performance.
5	Major defect leading to a loss of performance	The customer considers the defect to be major; repair that does not cost much must be carried out before standard service.	The failure leads to major deterioration in vehicle or equipments performance.
6		The customer considers the defect to be major; costly repair must be carried out before standard service. The defect leads to major deterioration in perceived quality, which has an impact on the way the customer views the brand image of RENAULT. Very slight injury not requiring hospitalization without incapacitating effects (Small cut ...)	The failure leads to major deterioration in vehicle or equipments performance.
7	Off-road breakdown or loss of confidence	The failure makes the driver stop the vehicle with the possibility of delaying the stop; the customer can no longer rely on the vehicle.	The customer feels that the failure makes it impossible to use the vehicle (e.g.: "very loud noise"), but the customer could use the vehicle in complete safety.
8		The failure immediately leads to an off-road breakdown (*the customer cannot delay the stop while driving*).	The failure makes it impossible to use the vehicle: e.g.: "does not start", "stopping the engine while driving".
9	Potentially related to safety or regulatory	The extreme nature of the consequences as regards the customer / users of the vehicle is relative to individuals' safety and/or to non-compliance with regulations.	Disturbed function leads to non-compliance with regulations or with an EIC-PS (Undesirable Customer Event that is "Potentially related to Safety"), with warning (1).
10		The extreme nature of the consequences as regards the customer / users of the vehicle is relative to individuals' safety safety and/or to non-compliance with regulations.	Disturbed function leads to non-compliance with regulations or with an EIC-PS (Undesirable Customer Event that is "Potentially related to Safety"), without warning (1).

(1) **Warning:**

- A change is perceived and is considered abnormal by one of the senses or unusual behaviour is noticed, the nature of which and/or intensity and/or development will be perceived without a doubt by most customer / users of the product/manufacturing/assembly as an anomaly that is likely to become a more serious undesirable event;
- This anomaly must encourage the customer to decide upon the appropriate instinctive action (2), in a reasonable frame of time that will allow the customer to execute this action for the safety of all users;
- The time it takes for the undesirable event to appear must be long enough;
- The time it takes for a more serious undesirable event to appear is given a value. The reproducible aspect of the warning is validated either by simulation or by physical tests, etc.

(2) **this implies that the designer would have identified beforehand the "appropriate instinctive" reactions that the customer would naturally adopt when a** warning **is noticed.**

Examples of warning, as information only, to evaluate the Severity of the effect on the customer / user of the product:	
Sight	oil leakage / smoke that the customer / user can see, warning light on the dashboard, misaligned steering wheel, etc.
Touch	vibration perceptible in steering wheel/pedals, abnormal gearshift lever oscillation, hot point, etc.
Smell	odours related to heat/fuel inside the cabin, etc.
Hearing	sound alert from the on-board computer, loud metal on metal sounds (e.g.: loose link that will not break in the short term), clanking/squeaking, etc.
Unusual behaviour	cold starting the vehicle becomes more and more difficult, right/left play on the wheel, degraded straight-ahead ability, erratic acceleration noticed, no response to a command or unusual response, etc.

For information purposes: table of equivalence between FMEA, HCPP and NISSAN Grade severities:

Severity in Renault FMEA	Severity in HCPP	NISSAN grade
10	**S or R**	**A**
9	**S or R**	**A**
8	**1**	**B1**
7	**1**	**B2**
6	**2**	**C**
5	**2**	**C**
4	**3**	**D**
3	**3**	**D**
2	**3 or 4***	**D or E**
1	**3 or 4***	

Note: FMEA Severities 1 and 2 correspond to HCCP Severity class 4 when the latter is used.

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S	Criteria to be used only in the Process FMEA: Effect that product non-compliance has on manufacturing/assembly
1	Product non-compliance leads to, at the most, a minor problem related to operation or for the operator.
2	Product non-compliance leads to minor facility failure; no noticeable deterioration of material: as information, Repair Time < 3 min. Production sequence is not disrupted. Products may be reworked on the line and at the workstation (less than 100% of production); no scrap.
3	Non-compliant products may be reworked on the line but not at the workstation (less than 100% of production); no scrap.
4	Non-compliance of the product leads to average facility failure. This requires short-term repair work of the facility. For information: 3 min < Repair Time < 20 min. Production sequence is slightly disrupted. Non-compliant products must be sorted. No scrap. A part (less than 100% of production) is reworked on the line but not at the workstation. Loss of production is negligible.
5	100% of production must be reworked, or the vehicle/unit must be repaired off line but is not sent to the repair workshop.
6	Product non-compliance leads to significant facility failure. This requires long-term service. For information: 20 min < Repair Time < 60 min. Production sequence is moderately disrupted. Less than 100% of production must be scrapped (sorting not necessary), or the vehicle/unit must be repaired in the repair workshop; the time taken for repair will be less than half an hour. Loss of production is moderate.
7	Less than 100% of production must be scrapped after sorting, or the vehicle/unit must be repaired in the repair workshop; the time taken for repair will be between half an hour and one hour.
8	Product non-compliance leads to serious facility failure or significant damages to material. For information: 60 min < Repair Time < 180 min. Production sequence is severely disrupted. 100% of production is scrapped or the vehicle/unit must be repaired in the repair workshop; the time taken for repair is greater than one hour. Loss of production is significant.
9	Product non-compliance leads to critical failure of the facility. This requires Repair Time > 180 min. The potential failure of the product may endanger manufacturing and maintenance personnel; with warning (1). ?
10	Product non-compliance brings production to a halt. The potential failure of the product may endanger manufacturing and maintenance personnel; without warning (1).

Examples of warning indicators, as information only, to evaluate the Severity of the effect product non-compliance has on manufacturing/assembly:	
Sight	oil leakage / smoke that the manufacturing/assembly personnel can see, warning light on the console, rotating light activated, etc.
Touch	unusual vibration, hot point, etc.
Smell	heat-related odours, chemical odours, etc.
Hearing	console sound alert, loud metal on metal sounds (e.g.: loose link that will not break in the short term), clanking/squeaking, air leak from HP pipe, etc.
Behaviour	robot stops in unusual position, etc.

Note: for every potential failure where an effect on manufacturing/assembly is mentioned, **the effect on the customer / user of the product will have to be mentioned and must appear in the FMEA**. Risks must be cleared in both cases.

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Design and Process FMEA – Severity Index S (summary table provided for information purposes)

Severity			General defect types	Criteria to be used in Design and Process FMEA:		Criteria to be used only in the Process FMEA:
FMEA	HCPP*	Nissan*		Effect on the customer / user of the product	Disturbed product service function	Effect that product non-compliance has on manufacturing/assembly
1	3 or 4		tolerated or not perceived by the	The customer does not notice the defect.	No obvious deterioration in vehicle or equipments performance.	Product non-compliance leads to, at the most, a minor problem related to operation or for the operator.
2		D or E E		The customer tolerates the defect.	No obvious deterioration in vehicle or equipments performance.	Product non-compliance leads to minor facility failure; no noticeable deterioration of material: as information, Repair Time < 3 min. Production sequence is not disrupted. Products may be reworked on the line and at the workstation (less than 100% of production); no scrap
3	3	D	Minor defect leading to a product approval comfort defect	The defect, which the customer considers to be minor, leads to a defect impacting the driving comfort; the cost of repair is very low and can wait until standard service.	The failure leads to a very minor deterioration in vehicle or equipments performance.	Non-compliant products may be reworked on the line but not at the workstation (less than 100% of production); no scrap.
4				The defect, which the customer considers to be minor, leads to a defect impacting the driving comfort; the cost of repair is low and can wait until standard service.	The failure leads to minor deterioration in vehicle or equipments performance.	Non-compliance of the product leads to average facility failure. This requires short-term repair work of the facility. For information: 3 min < Repair Time < 20 min. Production sequence is slightly disrupted. Non-compliant products must be sorted. No scrap. A part (less than 100% of production) is reworked on the line but not at the workstation. Loss of production is negligible.
5	2	C	Major defect leading to a loss of performance	The customer considers the defect to be major; repair that does not cost much must be carried out before standard service.	The failure leads to major deterioration in vehicle or equipments performance.	100% of production must be reworked, or the vehicle/unit must be repaired off line but is not sent to the repair workshop.
6				The customer considers the defect to be major; costly repair must be carried out before standard service. The defect leads to major deterioration in perceived quality, which has an impact on the way the customer views the brand image of RENAULT.	The failure leads to major deterioration in vehicle or equipments performance. Very slight injury not requiring hospitalization without incapacitating effects (Small cut ...)	Product non-compliance leads to significant facility failure. This requires long-term service. For information: 20 min < Repair Time < 60 min. Production sequence is moderately disrupted. Less than 100% of production must be scrapped (sorting not necessary), or the vehicle/unit must be repaired in the repair workshop; the time taken for repair will be less than half an hour. Loss of production is moderate.
7	1	B2	Off-road breakdown or loss of confidence	The failure makes the driver stop the vehicle with the possibility of delaying the stop; the customer can no longer rely on the vehicle.	The customer feels that the failure makes it impossible to use the vehicle (e.g.: "very loud noise"), but the customer could use the vehicle in complete safety.	Less than 100% of production must be scrapped after sorting, or the vehicle/unit must be repaired in the repair workshop; the time taken for repair will be between half an hour and one hour.
8		B1		The failure immediately leads to an off-road breakdown (*the customer cannot delay the stop while driving*).	The failure makes it impossible to use the vehicle: e.g.: "does not start", "stopping the engine while driving".	Product non-compliance leads to serious facility failure or significant damages to material. For information: 60 min < Repair Time < 180 min. Production sequence is severely disrupted. 100% of production is scrapped or the vehicle/unit must be repaired in the repair workshop; the time taken for repair is greater than one hour. Loss of production is significant.
9	S or R	A	Potentially related to safety	The extreme nature of the consequences as regards the customer / users of the vehicle is relative to individuals' safety and/or to non-compliance with regulations.	Disturbed function leads to non-compliance with regulations or with an EIC-PS (Undesirable Customer Event that is "Potentially related to Safety"), with warning (1).	Product non-compliance leads to critical failure of the facility. This requires Repair Time > 180 min. The potential failure of the product may endanger manufacturing and maintenance personnel; with warning (1). ?
10				The extreme nature of the consequences as regards the customer / users of the vehicle is relative to individuals' safety and/or to non-compliance with regulations.	Disturbed function leads to non-compliance with regulations or with an EIC-PS (Undesirable Customer Event that is "Potentially related to Safety"), without warning (1).	Product non-compliance brings production to a halt. The potential failure of the product may endanger manufacturing and maintenance personnel; without warning (1).

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Ranking tables (continued)

Ranking used in Design FMEA: Occurrence index O

Design FMEA: Occurrence index O				
O		• **Technical solution** • **Reliability** • **Functional characteristics**	**Environment, use, internal interactions or inter-system interactions**	**Probability P1 X P2/1**
Very low	**1**	• Known and robust solution • Proven and compliant reliability • Defined characteristics	Known, integrated with the technical solution and under control	[0 à 10.10^{-6}[
Low	**2**	• To a large extent known and robust solution • Reliable in relation to the purpose but proven in a slightly different environment • Defined characteristics	To a large extent known, integrated with the technical solution and under control	[10.10^{-6} à 100.10^{-6}[
	3			[100.10^{-6} à 500.10^{-6}[
Moderate	**4**	• The solution generally corresponds to vehicles in production • Reliability: failures appear occasionally in reduced numbers • Partially defined characteristics	Partially known or integrated with the technical solution	[500.10^{-6} à $1\,000.10^{-6}$[
	5			[$1\,000.10^{-6}$ à $2\,000.10^{-6}$[
	6			[$2\,000.10^{-6}$ à $5\,000.10^{-6}$[
High	**7**	• Not very known or robust solution • Reliability estimated but not in relation to the purpose • Characteristic: partially adapted	Very partially known or integrated with the technical solution	[$5\,000.10^{-6}$ à $10\,000.10^{-6}$[
	8			[$10\,000.10^{-6}$ à $20\,000.10^{-6}$[
Very high	**9**	• Solution, new or not robust • Unknown reliability • Characteristic: not defined or adapted	Unknown or not integrated with the technical solution	[$20\,000.10^{-6}$ à $50\,000.10^{-6}$[
	10			[$50\,000.10^{-6}$ à $100\,000.10^{-6}$]

Ranking tables (continued)

Ranking used in Process FMEA: Occurrence index O

PROCESS FMEA: Occurrence index O				
O		**P2/1: Probability that the product-cause (product non-compliance) that has appeared leads to a potential failure mode**	**P1: Probability of the product-cause appearing (product non-compliance due to the process)**	**P1 X P2/1**
Very low	**1**	Technical solution **insensitive** to the dispersion of product characteristics: **P2/1 tends towards 0** *The product-cause does not lead to failure mode.*	**Very low** probability of having a non-compliant product characteristic: **P1 tends towards 0** *Controlled process or procedure.* *Controlled Dispersion of production that is significantly low in relation to Tolerance.*	[0 à 10.10^{-6}[
Low	**2**	Technical solution **that is slightly sensitive** to the dispersion of product characteristics. *It is not very probable that the product-cause will lead to failure mode.*	**Low** probability of having a non-compliant product characteristic: *Controlled process or procedure with a margin, which can appear low.* *Controlled Dispersion of production that is significantly low in relation to Tolerance, but exceptional random defect.*	[10.10^{-6} à 100.10^{-6}[
	3			[100.10^{-6} à 500.10^{-6}[
Moderate	**4**	Technical solution **that is moderately sensitive** to the dispersion of product characteristics. *It is fairly probable that the product-cause will lead to failure mode.*	**Moderate** probability of having a non-compliant product characteristic. *Controlled process or procedure that is occasionally unstable.* *Controlled Dispersion of production but risk of unexpected drift.*	[500.10^{-6} à $1\ 000.10^{-6}$[
	5			[$1\ 000.10^{-6}$ à $2\ 000.10^{-6}$[
	6			[$2\ 000.10^{-6}$ à $5\ 000.10^{-6}$[
High	**7**	Technical solution **that is very sensitive** to the dispersion of product characteristics. *It is very probable that the product-cause will lead to failure mode.*	**High** probability of having a non-compliant product characteristic *Partially controlled process or procedure.* *Dispersion of production that is partially controlled and high in relation to Tolerance.*	[$5\ 000.10^{-6}$ à $10\ 000.10^{-6}$[
	8			[$10\ 000.10^{-6}$ à $20\ 000.10^{-6}$[
Very high	**9**	Technical solution that is **directly influenced** by dispersions of product characteristics: **P2/1 tends towards 1** *It is practically certain that the product-cause will lead to failure mode.*	**Very high** probability of having a non-compliant product characteristic: **P1 tends towards 1** *Uncontrolled or unstable process or procedure.* *Dispersion of uncontrolled production that is significantly high in relation to Tolerance.*	[$20\ 000.10^{-6}$ à $50\ 000.10^{-6}$[
	10			[$50\ 000.10^{-6}$ à $100\ 000.10^{-6}$]

Note*: depending on the analysis ratio as regards qualitative evaluation, take the highest corresponding O index. As regards quantitative evaluation, take the corresponding O index.*

To estimate the Occurrence index O in the Process FMEA, the indicative values given below can also be used:

Process FMEA: Occurrence index O if P2/1 is considered to be equal to 1	
O	**Indicative values of centered Cpk[1]**
1	[∞ à 1,46[
2	[1,46 à 1,30[
3	[1,30 à 1,16[
4	[1,16 à 1,10[
5	[1,10 à 1,03[
6	[1,03 à 0,93[
7	[0,93 à 0,86[
8	[0,86 à 0,77[
9	[0,77 à 0,65[
10	[0,65 à 0]

[1] Centered Cpk => Cap = Tolerance / Dispersion and no off centering*: P1*P2/1 can be directly deducted from this value **when P2/1 = 1**

*for centered distribution only: in that case, mean of the sample equals to nominal value.

Ranking tables (continued)

Ranking used in Design FMEA: Detection index D

Design FMEA: Detection index D		
D	**Detection is the action planned in order to prevent the potential cause and/or failure mode that is supposed to have appeared from affecting task N+1.**	**Probability P3**
1	**No potential failure will affect task n+1.** *Examples:* • *the product validation process is complete and appropriate as regards S/R (Safety/Regulation) requirements* • *tests related to the component, system, unit, vehicle, etc. are representative of usage by customers and the product is representative of mass production* • *opinions of Technical experts are convergent* • *sequence of planned Detection is optimal (calculation, simulation, prototypes, etc.)*	[0 à 0,1%[
2	**It is not very probable that a potential failure affects task n+1**. *Examples*: • *the product validation process is complete and appropriate to a large extent as regards S/R requirements* • *tests related to the component, system, unit, vehicle, etc. are representative of usage* • *digital simulation or calculation methods are tested (constraint-resistance diagram, etc.)* • *provisions taken are in compliance with Technical department standards* • *the sequence of Detections is practically optimal, etc.*	[0,1 à 0,2%[
3		[0,2 à 0,3%[
4	**It is slightly probable that a potential failure affects task n+1**. *Examples*: • *the product validation process is partial and does not integrate all variables as regards S/R requirements* • *tests related to the component, system, unit, vehicle, etc are partially representative of usage (all elements related to the environment are not integrated)* • *digital simulation or calculation methods are partial (dynamic aspects, chain of dimensions in 1 dimension instead of 2 dimensions, etc. not considered)* • *the Technical department standard does not integrate all parameters* • *the sequence of planned Detection is partial, etc.*	[0,3 à 0,5%[
5		[0,5 à 1%[
6		[1 à 2%[
7	**It is fairly probable that a potential failure affects task n+1**. *Examples*: • *the product validation process is not very appropriate as regards S/R requirements* • *tests related to the component, system, unit, vehicle, etc are not really representative of usage (no integration from the environment, etc.)* • *calculation or digital simulation is not totally appropriate (in nominal configuration, etc.)* • *drawing review, etc.*	[2 à 3%[
8		[3 à 5%[
9	**It is highly probable that a potential failure affects task n+1**. *Examples*: • *the product validation process is not appropriate as regards S/R requirements or it is not planned at all* • *tests related to the component, system, unit, vehicle, etc. are not representative* • *calculations or digital simulations are not appropriate or are not planned* • *detection is not planned or is not effective*	[5 à 10%[
10		[10 à 100%]

Ranking tables (continued)

Ranking used in Process FMEA: Detection index D

Process FMEA: Detection index D		
D	**Detection is the action planned in order to prevent the potential cause and/or failure mode that is supposed to have appeared from affecting task N+1.**	**P3**
1	**The failure has a very low probability of affecting task N+1: P3 => 0** The different inspection procedures or the events that make it impossible to continue the manufacturing or assembly process can be expected to detect the failure with certainty. *Examples*: • *100% automatic inspection + process for calibration and isolating non-compliance in an efficient manner* • *POKA-YOKE, where it is absolutely impossible to continue the manufacturing process* • *functional characteristic that is obviously detectable (no steering wheel, vehicle non-ignition, etc.)*	[0 à 0,1%[
2	**The failure has a low probability of affecting task N+1:** The different inspection procedures or the events that make it impossible to continue the manufacturing or assembly process can probably be expected to detect the failure. *Examples*: • *statistical product or process control + maintenance plan of the installation* • *simple unit inspection by an operator* • *the nature of the impossibility to make the assembly can let a few failures pass* • *the fault locator may let a few failures pass, etc.*	[0,1 à 0,2%[
3		[0,2 à 0,3%[
4	**The failure has a moderate probability of affecting task N+1:** The different inspection procedures or the events that make it impossible to continue the manufacturing or assembly process do not detect the failure effectively. *Examples:* • *inspection with an appropriate frequency, with production monitoring.* • *100% manual inspection where Detection is difficult (measure, appearance, etc.).* • *the manufacturing or assembly process allows defects to pass rather easily, etc.*	[0,3 à 0,5%[
5		[0,5 à 1%[
6		[1 à 2%[
7	**The failure has a high probability of affecting task N+1:** The different inspection procedures or the events that make it impossible to continue the manufacturing or assembly process do not detect the failure. *Examples*: • *control by sampling, incorrectly adapted* • *subjective control of the non-compliant characteristic* • *non-conformity that cannot be easily detected* • *ineffective detection: inspection might work correctly, but non-compliant parts run the risk of being mixed with compliant parts, etc.*	[2 à 3%[
8		[3 à 5%[
9	**The failure has a very high probability of affecting task N+1: P3 => 1** The defect is not detected during manufacturing, assembly or control. *Examples*: • *detection deals with a potential failure mode, which affects the life cycle of the component (mode cannot be detected at 0 km)* • *the defect cannot be detected easily (incomplete electrical connection, visual inspection in a difficult environment, etc.)* • *the characteristic that is considered to be non-compliant is not controlled* • *the non-compliant characteristic that is controlled is not identified as being non-compliant*	[5 à 10%[
10		[10 à 100%]

APPENDIX 5: NORMATIVE: ADDITIONAL FLOW FMEA

The FLOW FMEA is designed in the same way as the Design and Process FMEA.

The most significant differences are defined below. Supplementary information to the FMEA standard is expected.

Potential EFFECT of failure

In the flow FMEA, **the manufacturer and central production departments** (DLI) (see below) **are the Customers**.

DETECTION

Detection is the action planned in order to prevent the cause (and/or mode) from affecting the manufacturer and the Industrial Logistics Department.

RANKING TABLE

Flow FMEA: Ranking varies from 1 to 4 for O and D and from 1 to 5 for S (see below).

ORGANISING INTO A HIERARCHY AND DECISION-MAKING THRESHOLDS

To obtain an overall image of the level of risk of the studied system, the RPNs are represented in the form of a bar graph.

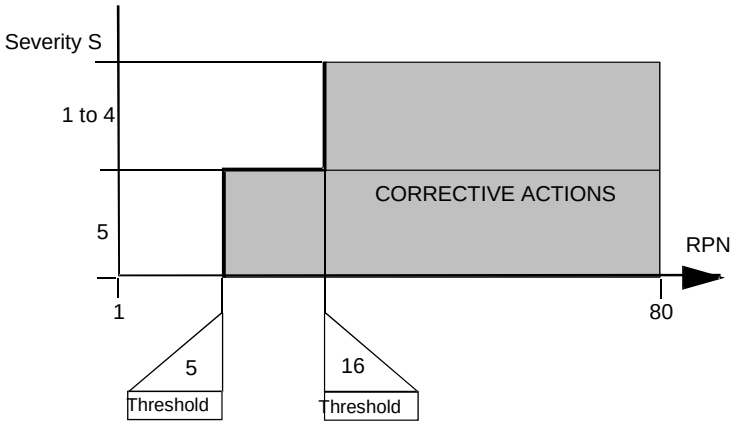

Figure 10: Flow FMEA

Flow FMEA file

Compile a file and archive it, in paper or electronic format. It will include the following:

	Flow FMEA
Initialisation sheet.	X
Data from the studied subject and modifications:	
- operating modes.	X
Functional analysis:	
- study on functions,	X
- flows diagram.	X
Sheets related to analyses, evaluation and corrective actions.	X
Summary notes.	X

Ranking tables

Ranking used in Flow FMEA: Severity index S

S	Consequences for users			
		Product **and/**	**or** Volume **and/**	**or** Order
1	No consequence	OK	OK	OK
2	Annoyed	Non-compliant; rework on the line	Small loss of volume	Order slightly disturbed (indicator of vehicles in the cycle (1))
3	Dissatisfied	Non-compliant; rework off the line	Moderate loss of volume	Order moderately disturbed (indicator of vehicles in the cycle (1))
4	Very dissatisfied	The non-compliance affects the internal Customer (a manufacturing sector that is downstream from the studied process)	Significant loss of volume	Order significantly disturbed (indicator of vehicles in the cycle (1))
5	Catastrophic	The non-compliance may affect the customer / user of the product	Production stop	

(1) The loss of volume % and vehicles-in-cycle indicator are to be defined at the beginning of the FMEA by the manufacturer and the Central Production Department (DPLI).

Occurrence index O:

O	**Probability of appearance (P1 x P2/1)** Note: in the absence of P1 and P2/1, appearance is stated in number of days.
1	Less than once per month
2	Once per month Once in [6 to 20 days]
3	Once per week Once in [2 to 5 days]
4	Once or n times per day

Detection index D:

D	**Probability P3 where the cause or mode may affect a downstream Manufacturing sector**				
	P3	**Practical provision**			
		Flow level measure	**Alert**	**Reactivity**	**Measure**
1	Effective detection P3 tends towards 0	YES	YES	YES	YES
2	Moderately effective	YES	YES	YES	NO
3	Slightly effective	YES	YES	NO	NO
4	P3 tends towards 1 Detection is ineffective or no Detection	NO	NO	NO	NO

CHOICE OF SUBJECTS TO BE DEALT WITH IN THE FLOW FMEA:

	STATUS	PERFORMANCE (volume to be produced, production order, product compliance)	IMPLEMENTATION OF A FLOW FMEA	
			Decision	Objective
F L O W	Carryover of an existing Operating mode (for a future vehicle, a future unit or a new production facilities)	Known and compliant to the objective	NO	
		Unknown	YES	To improve the planned operating mode.
	New operating mode		YES	To validate the planned operating mode.
	Modified operating mode		YES	To analyse the modified part.
	Mode that runs the risk of production being blocked		YES	To analyse the part that runs the potential risk.

LISTING SOURCE DATA AND FUNCTIONAL ANALYSIS TO BE IMLEMENTED BY A TYPE OF FMEA:

INPUT DATA	TYPE OF ANALYSIS TO BE USED	FLOW FMEA
Operating mode	Functional analysis of flows and Use existing flow diagram or draw it if one does not exist	YES YES

APPENDIX 6.1: FOR INFORMATION PURPOSES: DESIGN FMEA DIAGNOSIS GRID

Vehicle or project: Unit:
Subject:
Requestor (Name, department, company):
Coordinator:

	KEY POINTS	YES	NO	RECOMMENDATION (in case the response is no)
	SUMMARY SHEET: DATA			
1	**1. Deadlines are met:** In relation to the milestone date, (TGA) analysis, follow-up and verification dates are observed. **2. The reasons for creating the FMEA are defined:** Product: innovative, prone to risk, new, etc.			
	ANALYSIS			
2	**In relation to planned limits, the processed perimeter is complete:** Design FMEA: - functions - components - product characteristics, etc. are integrated			
3	**1. Input data is quantified:** - service functions: handling force 15 daN ± 2, etc. - component Dimensions 20 ± 0,1, material, etc. **2. Quantified input data is included in documentation related to potential failures**: - cause: dimensions > 20.1; dimensions < 19.9, etc. - mode: part breaks after 100000 kilometres, etc. - Detection: durability testing No. 5.5. 1944, ...			
4	**Analysis of potential failures tends towards completeness:** - There are for ex. from 5 to 7 modes for a given function - There are for ex. from 5 to 7 product causes for a given mode - Every mode has an effect on the customer-motorist - Detection is documented for every cause-mode pair			
	EVALUATION			
5	**The Renault evaluation table is used or the ranking table that is applied is correlated:** - The Severity (S) index is adapted to the customer / user-effect - Occurrence index (O): • cause-mode pairs are evaluated - Detection index • all Detection is evaluated			
	DECISION-MAKING: ACTION PLAN			
6	**"Who will do what and when?" is defined:** Reminder about thresholds: S = 9 and 10 : RPN ≤ 36 S = 5 to 8 : RPN ≤ 50 S = 1 to 4 : RPN ≤ 100 With regard to RPNs that exceed the threshold: 1. corrective measures to be studied are put forward 2. manager of the resolution is identified 3. Target Completion Date for presenting the corrective solution is defined			
	FOLLOWING UP CORRECTIVE SOLUTIONS			
7	**1 – On the planned date, corrective solutions that have been formalised are presented:** - product and/or modified characteristics - Created or improved detection, calculation results, testing are in accordance with the objective			

	- all documents are present **2 – New RPNs are calculated after corrective solutions have been analysed and evaluated:** 1. the follow-up date is given on the initialization sheet 2. analysis contents are modified 3. Severity (S) index does not change (else, explain). 4. Occurrence (O) and Detection (D) indexes change according to the contents of the analysis 5. RPNs that exceed thresholds are resolved			

VERIFY RESULTS

8	**Potential failures are cleared:** 1. a summary report certifies whether the corrective solutions that have been set up and are effective 2. when the FMEA requestor signs at the bottom of the initialization sheet, s/he certifies that the potential failures have been cleared			

Evaluation grid of the Design FMEA

Analysis and evaluation	**Action plan**		**Results at TGA/RO/IOD**		
Analysis and prioritisation of failures	Planned before TGA/RO/IOD	All RPN > threshold are included	Results shown for all RPN > threshold	Results applied to the product and proven validation	Indicator (put a cross in the box that corresponds to the result)
OK	**OK**	**OK**	**OK**	**OK**	**K - 0**
Non-compliance exists, but making the product compliant is compatible with project time constraints.					**K - 10**
Not carried out	**Action plan not defined or incomplete and/or Target Completion Dates are incompatible with project constraints**		**One or several RPNs > threshold still remain, or all corrective actions have not been applied. Incompatible timelines**		**K - 50**

DECISION-MAKING

K - 0 Design FMEA study accepted

K - 10 Design FMEA study accepted subject to making the product compliant

K - 50 Design FMEA study rejected. Start an action plan.

REMARKS

Name: ..... Date: ..... Signature:

APPENDIX 6.2: FOR INFORMATION PURPOSES: PROCESS FMEA DIAGNOSIS GRID

Vehicle or project: Unit:
Subject:
Requestor (Name, department, company):
Coordinator:

	KEY POINTS	YES	NO	RECOMMENDATION (in case the response is negative)
	SUMMARY SHEET: DATA			
1	**1. Deadlines are met:** In relation to the milestone date, (TGA/RO/IOD and ABVC/AMS-lot or ABPT/AMEI) analysis, follow-up and verification dates are met. **2. The reasons for creating the Process FMEA are defined:** Process: innovative, prone to risk, rate of non-compliance, PPM to be reduced, etc.			
	ANALYSIS			
2	**In relation to planned limits, the processed perimeter is complete:** Process FMEA: - operations, inter-operations - planned rework - device for inspection, etc. are analysed			
3	**1. Input data is quantified:** - service functions: handling force 15 daN ± 2, etc. - operation: clamping torque 21 Nm ± 15 %, manufactured dimension: 20 ± 0,1, manufacturing parameters: temperature 200 ° ± 15°… **2. Quantified input data is included in documentation related to potential failures**: - product cause: dimensions > 20.1; dimensions < 19.9, etc. - mode: part breaks after 100000 kilometres, etc. - Detection: monitoring plan No. 5.5. 1944, Occurrence, ...			
4	**Analysis of potential failures tends towards completeness:** - there are for ex. from 5 to 7 modes for a given function - there are for ex. from 5 to 7 product causes for a given mode - there are for ex. from 5 to 7 process-causes for every product-cause (product non-compliance that is supposed to have appeared) - every mode has an <u>effect on the customer-motorist</u> - detection is documented for every cause-mode pair			
	EVALUATION			
5	**The Renault ranking is used or the ranking table that is applied is correlated:** - the effect on the customer / user of the product has to be indicated; manufacturing/assembly effect on the customer may be indicated - in both cases, the Severity index S is adapted to the customer-effect - Occurrence index (O): • cause-mode pairs are evaluated • P1 probabilities (and P2/1, preferably) are put forward - Detection index (D): • all Detection is evaluated • P3 probabilities are put forward preferably - all RPNs are calculated			
	DECISION-MAKING: ACTION PLAN			
6	**"Who will do what and when?" is defined:** Reminder about thresholds: S = 9 and 10 : RPN ≤ 36 S = 5 to 8: RPN ≤ 50 S = 1 to 4: RPN ≤ 100 With regard to RPNs that exceed the threshold: 1. corrective measures to be studied are put forward 2. manager of the resolution is identified 3. Target Completion Date for presenting the corrective solution is			

	defined			
	FOLLOWING UP CORRECTIVE SOLUTIONS			
7	**1. On the planned date, corrective solutions that have been formalised are presented:** - modified manufacturing process plan, procedure, facilities, workstation layout, etc. - improved product, process monitoring plan, inspection equipment and responsiveness, etc. - product is modified, if necessary **2. New RPNs are calculated after corrective solutions have been analysed and evaluated:** 1. the follow-up date is given on the initialization sheet 2. analysis contents are modified 3. Severity (S) index does not change (else, explain). 4. Occurrence (O, P1, P2/1) and Detection (D, P3) indexes change according to the contents of the analysis 5. RPNs that exceed thresholds are resolved			
	VERIFY RESULTS			
8	**Potential failures are cleared:** 1. a summary report certifies whether the corrective solutions that have been set up and are effective 2. when the FMEA requestor signs at the bottom of the initialization sheet, s/he certifies that the potential failures have been cleared			

Evaluation grid of the Process FMEA

Analysis and evaluation	**Action plan at TGA/RO/IOD**		**Result at ABVC/AMS-Lot or ABPT/AMEI**		
Analysis and prioritisation of failures	The action plan is planned	All RPN > threshold are included	Results shown for all RPN > threshold	Results applied to the process and monitoring plan	Indicator (put a cross in the box that corresponds to the result)
OK	**OK**	**OK**	**OK**	**OK**	**K - 0**
Non-compliance exists, but making the product compliant is compatible with project time constraints.					**K - 10**
Not carried out	**Action plan not defined or incomplete and/or Target Completion Dates are incompatible with project constraints**		**One or several RPNs > threshold still remain, or all corrective actions have not been applied. Incompatible timelines**		**K - 50**

DECISION-MAKING

K - 0 Process FMEA study accepted

K - 10 Process FMEA study accepted subject to making the product compliant

K - 50 Process FMEA study rejected. Start an action plan.

REMARKS

Name: ..... Date: ..... Signature:

APPENDIX 7: NORMATIVE: SUPPLIER FMEA SYNTHESIS TEMPLATE

GROUPE RENAULT

SUPPLIER DESIGN/PROCESS FMEA SYNTHESIS

Project:	Safety ☐ Renault 1 ☐ Regulation ☐ 2 ☐	Doc Ref. No. / Ver. Doc. Rev. Date: Doc. Origin Date:
Supplier Name: Supplier Plant Name:	Supplier Code: Author:	E-mail: Tel. No:
Part Name: Part No. & Issue Level: Design Note No. / DEVO:	Remark	

1	Limite of the FMEA study :	
2	FMEA Standard considered by the supplier (Internal, AIAG/VDA; ...) :	
3	RPN / AP(1) threshold considered for corrective mesures :	
4	UCEps(2) identified by the supplier FMEA :	
5	Liste of critical failures (Severity 8-9-10) identified with corresponding RPN / AP and realized corrective mesures : Fill the table below	

Severity (8 – 9 – 10)	Part / system Failure	Client Effect	RPN / AP (H & M) brefore corrective mesures	Realized and validated corrective mesures (above RPN treshold / AP H & M)	Final RPN / AP	Status (Open / Clos)

AP(1) : Action Priority following AIAG/VDA FMEA HANDBOOK | (2)UCEps : Unwated Custumer Event Pontentialy Safety - Refer to AEK-D-00051

Supplier Sign-off

	Project Leader	Purch / Sales	Production Control	Quality Assurance	Engineering	Design	Production				
Name											
Signature											

The Appendix 7, Supplier FMEA Synthesis Template in Excel format is attached to this PDF